

Fundamentals of ICT: Components of Computer System

Part A. Components of Computer System

I. History and Evolution of Computers The evolution of computers is generally tracked through "generations," each characterized by a major technological development that fundamentally changed how computers operate.

- **First Generation (1940–1956): Vacuum Tubes**

Technology: Used vacuum tubes for circuitry and magnetic drums for memory.

Characteristics: Enormous in size (taking up entire rooms), expensive to operate, generated huge heat, and relied on machine language (binary).

Examples: ENIAC (Electronic Numerical Integrator and Computer), UNIVAC.

- **Second Generation (1956–1963): Transistors**

Technology: Transistors replaced vacuum tubes.

Characteristics: Computers became smaller, faster, cheaper, and more energy-efficient. This era introduced Assembly language (using codes like "ADD" instead of binary).

Examples: IBM 7094, CDC 1604.

- **Third Generation (1964–1971): Integrated Circuits (ICs)**

Technology: Transistors were miniaturized and placed on silicon chips (semiconductors).

Characteristics: drastic increase in speed and efficiency. Users interacted via keyboards and monitors rather than punch cards.

Examples: IBM 360, PDP-11.

- **Fourth Generation (1971–Present): Microprocessors**

Technology: Thousands of integrated circuits were built onto a single silicon chip.

Characteristics: The birth of the Personal Computer (PC). Computers became small enough to fit on a desk (and later in a pocket). Introduction of GUIs (Graphical User Interfaces) and the Mouse.

Examples: Apple Macintosh, IBM PC.

- **Fifth Generation (Present & Beyond): Artificial Intelligence**

Technology: Based on Artificial Intelligence (AI), ULSI (Ultra Large-Scale Integration), and Parallel Processing.

Characteristics: Devices capable of learning, self-organization, and natural language processing.

Examples: IBM Watson, Siri/Alexa, Generative AI (like ChatGPT).

II. Classification of Computers Computers are classified based on their data handling capabilities, size, and utility.

- **By Operation Principle:**

Analog Computers: Process continuous data (e.g., Speedometer, Thermometer).

Digital Computers: Process discrete data (0s and 1s) with high accuracy (e.g., Laptops, Smartphones).

Hybrid Computers: Combine features of both (e.g., ECG machines in hospitals).

- **By Size and Power:**

Supercomputers: The most powerful systems used for complex calculations like weather forecasting and nuclear simulations (e.g., Fugaku, Frontier).

Mainframe Computers: Large systems supporting thousands of users simultaneously; used by large organizations for bulk data processing (e.g., Banking systems).

Minicomputers: Mid-range servers for medium-sized businesses; largely replaced by modern servers.

Microcomputers: Small, affordable computers designed for individual use (e.g., Desktops, Laptops, Tablets, Wearables).

III. Computer Hardware Components Hardware represents the physical, tangible parts of the computer system. It is divided into four main categories based on function.

- **Input Devices:** Hardware used to enter data and instructions into the computer.

Examples: Keyboard, Mouse, Scanner, Microphone, Webcam.

- **Processing Devices (System Unit):** The internal components that process data.

CPU (Central Processing Unit): The "brain" that executes instructions.

Motherboard: The main circuit board connecting all components.

RAM (Random Access Memory): Temporary workspace for active tasks.

- **Output Devices:** Hardware that displays the processed information to the user.

Examples: Monitor (Visual), Printer (Physical), Speakers (Audio), Projector.

- **Storage Devices:** Hardware used to store data permanently.

Examples: Hard Disk Drive (HDD), Solid State Drive (SSD), USB Flash Drives, Cloud Storage (Remote servers).

IV. Computer Software Systems Software is the set of intangible instructions or programs that tell the hardware what to do. It is divided into two main types.

- **System Software:**

Definition: Software designed to run the computer's hardware and provide a platform for other software to run. It acts as the interface between the hardware, the application, and the user.

Operating Systems (OS): Windows, macOS, Linux, Android, iOS.

Utility Programs: Antivirus, File Management, Disk Cleanup.

Device Drivers: Programs that help the OS communicate with hardware (e.g., a printer driver).

- **Application Software:**

Definition: Programs designed to help the user perform specific tasks or solve problems.

General Purpose: Word Processors (MS Word), Spreadsheets (Excel), Web Browsers (Chrome).

Specialized: Graphic Design (Photoshop), Accounting Software, Gaming.